



The impact of a mixture of medicinal herbs on ruminal fermentation, parasitological status and hematological parameters of the lambs experimentally infected with *Haemonchus contortus*



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ABSTRACT

The aim of this study was to determine the effect of dietary supplementation with a mixture of selected medicinal herbs (Herbmix) on hematological and parasitological parameters and on the inflammatory response of lambs infected experimentally with the gastrointestinal parasitic nematode *Haemonchus contortus*. Herbmix contained selected medicinal herbs common in central Europe (8.55% each of *Althaea officinalis*, *Petasites hybridus*, *Inula helenium*, *Malva sylvestris*, *Chamomilla recutita*, *Plantago lanceolata*, *Rosmarinus officinalis*, *Solidago virgaurea*, *Fumaria officinalis*, *Hyssopus officinalis* and *Melissa officinalis*, 5% *Foeniculum vulgare* and 1% *Artemisia absinthium*). Herbmix (100 g dry matter (DM)/d) was added to the basal diets of meadow hay (600 g DM/d) and a commercial concentrate (350 g DM/d) per animal. Twenty-four female lambs were divided into four groups: infected animals (Inf), infected animals supplemented with Herbmix (Inf + Herb), uninfected control animals (C) and uninfected animals supplemented with Herbmix (C + Herb). Inf and Inf + Herb lambs were infected orally with approximately 5000 L3 larvae of a strain of *H. contortus* susceptible to anthelmintics (MHco1). Herbmix supplementation for Inf + Herb and C + Herb began on day 0 (D0). Feces were collected and eggs per gram (EPG) of feces were quantified on D0, D20, D32, D50 and D60. Variables differed after 24 h of ruminal incubation of the individual medicinal herbs ($P < 0.001$), but the herbs had no negative effect on the fermentation patterns. The mean EPG counts were significantly lower by 45% between D32 and D50 ($P < 0.05$) and by 63% between D32 and D60 ($P < 0.01$) for the infected lambs receiving Herbmix (Inf + Herb). These EPG reductions were only 10 and 30% in infected group (Inf, $P > 0.05$). Mean cumulative gain in live weight was higher in Inf + Herb as compared to Inf by D32 ($P < 0.01$) and D50 ($P < 0.001$). The evaluation of red blood cells identified clinical signs of haemonchosis, such as anemia, in Inf and Inf + Herb. Total antioxidant capacity ($P < 0.008$) and activity of glutathione peroxidase ($P < 0.005$) in serum at D50 were lower in Inf and Inf + Herb than C and C + Herb. The concentration of serum calprotectin differed significantly ($P < 0.001$) in Inf vs. Inf + Herb. The Zn and Fe serum concentrations differed significantly ($P < 0.05$ and $P < 0.001$, respectively) in Inf vs. Inf + Herb, Inf vs. C and Inf vs. C + Herb by D32. The mixture of the medicinal herbs thus contributed to slowing the dynamics of infection and improving the production indicators of the animals.

1. Introduction

Maintaining high productivity and profitability in pasture-based small ruminants is hindered by infection with the gastrointestinal parasitic nematode *Haemonchus contortus*. Pastures provide a link between the free-living and parasitic phases of helminthic parasites of grazing animals, and species of herbs in the pasture may facilitate or prevent the survival of free-living populations and reduce or intensify

the effects of parasitism on the host (Waller et al., 2001; Hoste et al., 2011). *H. contortus* is mainly controlled by chemoprophylaxis by the repeated application of anthelmintics. The current global trend in the production of ecologically clean foods, however, has increased the demand for organic products, because the repeated use of anthelmintics increases the risk of residues in food products and the development of anthelmintic resistance. The use of medicinal herbs containing plant secondary metabolites with bioactive properties is an alternative

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strategy to control nematode populations. Medicinal herbs can affect a population of parasites at various stages of their life cycle (Waller and Thamsborg, 2004; Yadav et al., 2010; Wink, 2012).

Selected medicinal herbs or herbal extracts containing tannins, essential oils, saponins, flavonoids, polyphenols and other secondary metabolites may also improve ruminal fermentation and metabolism (Patra and Saxena, 2009; Bhatta et al., 2013; Wencelová et al., 2015). These phytogetic additives enhance a variety of important processes in the animal body and positively affect feed quality, the health of animals and animal products by their specifically efficacious substances that improve the nutritional condition of animals and can also increase host immunity against parasites (Patra, 2007; Karaskova et al., 2015). Direct antiparasitic effects of herbs or herbal extracts on parasites have long been known (Hammond et al., 1997; Coop and Kyriazakis, 2001; Hoste et al., 2006), and forages containing condensed tannins especially provide sheep with the ability to resist parasitic infection (Niezen et al., 1995). Phytotherapeutic treatments in traditional medicine (Viegi et al., 2003; Lans et al., 2007) combined with the pharmacology of biologically active compounds (Githiori et al., 2006) can be used in veterinary medicine to control gastrointestinal helminths of small ruminants. Ruminants may also learn to self-medicate against gastrointestinal parasites by increasing the consumption of plant secondary compounds with antiparasitic actions (Villalba et al., 2014). Feed rations supplemented with a dry mixture of selected herbs with deworming properties can simulate natural grazing conditions on pastures with high diversity of herbs and improve the health of animals with an effective control of gastrointestinal parasites.

Mineral and antioxidant status are directly related with susceptibility of animals to the parasite infection. Imbalance in mineral status results in prolonged duration and increased intensity of parasite infection. Zinc and iron are essential for the improvement of the immune functions because the efficiency of immunological responses, mainly intestinal immunity, against gastrointestinal nematodes depends on the zinc status and Zn nutrition (Hughes and Kelly, 2006). The nematode infection may induce production of reactive oxygen species, which may damage the parasites but generate oxidative stress (Machado et al., 2014). Medicinal plants rich in polyphenols can represent interesting natural antioxidants that could be used in animal nutrition to protect from the potentially damaging oxidative damage. It has been hypothesized that plant antioxidants may contribute to the beneficial health effects of dietary medicinal herbs and provide a better defense against the oxidative stress of lambs caused by *H. contortus* infection.

The selection of 13 traditional medicinal herbs (Herbmix) in this study was based on their antiparasitic (Waller et al., 2001; Viegi et al., 2003; Mali and Mehta, 2008), anti-inflammatory, carminative (Kresánek and Kresánek, 2008), antioxidant and immunological (Hosoda et al., 2006; Fraisse et al., 2007) properties. Our goal was to investigate (1) the effect of individual dry medicinal herbs on ruminal fermentation in vitro, (2) the effect of Herbmix (as a dietary component) on parasitological status and mean gain in the live weights of lambs experimentally infected with *H. contortus* and (3) the effect of Herbmix on the hematological, immunological and antioxidative parameters and on the mineral status of the infected lambs.

2. Materials and methods

2.1. Twenty-four hour in vitro batch-culture experiment of Herbmix

Each herb of the Herbmix mixture (non-commercial product) was incubated in vitro in ruminal fluid to assess its effect on parameters of ruminal fermentation (pH, ammonia, gas, methane and short-chain fatty acids (SCFAs)). The ruminal fluid was collected from the thoroughly mixed ruminal contents of slaughtered lambs, packed in pre-warmed flasks and transported to the laboratory. The inoculum was passed through four layers of gauze, mixed at a 1:1 ratio with

McDougall's buffer (McDougall, 1948) and purged with CO₂. The inoculum was dispensed in volumes of 35 ml into serum bottles (120 ml) containing 0.25 g of dried herb. The dry herbs were obtained from commercial sources (AGROKARPATY, Plavnica, Slovak Republic and BYLINY Mikeš s.r.o., Čičenice, Czech Republic): roots of marsh-mallow (*Althaea officinalis* L.), butterbur (*Petasites hybridus* L.) and elecampane (*Inula helenium* L.); flowers of mallow (*Malva sylvestris* L.) and chamomile (*Chamomilla recutita* L.); leaves of ribwort plantain (*Plantago lanceolata* L.) and rosemary (*Rosmarinus officinalis* L.); seeds of fennel (*Foeniculum vulgare* Mill.) and stems of goldenrod (*Solidago virgaurea* L.), fumitory (*Fumaria officinalis* L.), hyssop (*Hyssopus officinalis* L.), lemon balm (*Melissa officinalis* L.) and wormwood (*Artemisia absinthium* L.). The herbs were ground through a 0.15–0.40 mm screen using a grinder (Molina, MIPAM, České Budějovice, Czech Republic) or standmixer (ASSISTANTM 2600, Berlin, Germany) and fermented in vitro with buffered ruminal fluid. The fermentation bottles were filled with CO₂, closed with a butyl rubber stopper, sealed with aluminum crimp caps and incubated for 24 h at 39 ± 0.5 °C with periodic mixing. Batch cultures of ruminal fluid were incubated as described elsewhere (Váradyová et al., 2005; Wencelová et al., 2014). The in vitro experiment consisted of a fermentation with each of the thirteen herbs as the sole substrate. Three replicates (three incubation bottles) were prepared for each herb, and the experiment was repeated three times within three consecutive days (n = 3 × 3). Three replicate bottles were also used for the blank (ruminal inoculum, no substrate).

2.2. Animals, diets and experimental design

Animal use and study design were approved by the Ethics Committee of the Institute of Parasitology of the Slovak Academy of Sciences in accordance with the national legislation in Slovakia – Animal Welfare Act No. 23/2009. Twenty-four female lambs (Improved Valachian, 3–4 months of age, initial body weights of 12.5 ± 1.64 kg) were housed in common stalls for 15 d for acclimatization to feeding, with free access to water. Each animal was fed meadow hay (600 g DM/d) and a commercial concentrate (350 g DM/d; composed of 70% barley, 22% soybean meal, 4.8% wheat bran, 0.5% bicarbonate and 2.7% mineral–vitamin premix). The lambs were then randomly divided into four groups (n = 6/group, one stall per group): animals infected by *H. contortus* (Inf), infected animals supplemented with Herbmix (Inf + Herb), uninfected control animals (C) and uninfected animals supplemented with Herbmix (C + Herb). Inf and Inf + Herb lambs were infected orally with approximately 5000 L3 larvae of a strain of *H. contortus* susceptible to anthelmintics (MHco1). Inf + Herb and C + Herb lambs were fed Herbmix (100 g DM/d/animal) beginning from day 0 (D0). Herbmix contained 8.55% each of *A. officinalis*, *P. hybridus*, *I. helenium*, *M. sylvestris*, *C. recutita*, *P. lanceolata*, *R. officinalis*, *S. virgaurea*, *F. officinalis*, *H. officinalis* and *M. officinalis*, 5% *F. vulgare* and 1% *A. absinthium*. Herbmix was stable throughout the experiment, daily mixed with commercial concentrate and supplemented from D0 to D60. The chemical compositions of the meadow hay, commercial concentrate, individual medicinal herbs and Herbmix are given in Table 1. The lambs were weighed on D0, D20, D32 and D50.

2.3. Measurements and chemical analysis of dietary substrates

The dietary substrates (individual medicinal herbs, Herbmix, meadow hay and commercial concentrate) were analyzed in triplicate by standard procedures (AOAC, 1990). The DM content was obtained by drying the samples at 105 °C for at least 24 h in an oven (method no. 930.15). The total ash content of the samples was determined by ashing overnight at 550 °C (method no. 942.05) in a muffle furnace. Nitrogen (N) content (method no. 968.06) was determined using a FLASH 4000 (Thermo Fisher Scientific, Cambridge, UK). Crude-protein content was calculated by multiplying the total N content by 6.25 (method no. 990.03). The acidic-detergent fiber (ADF) and neutral-detergent fiber

Table 1

Dry matter (g/kg), chemical composition, in vitro dry-matter digestibility (g/kg dry matter, DM; n = 3) and qualitative phytochemical screening of the experimental herbs.

Plant species	Plant family	Part	DM	aNDF	ADF	CP	N	Ash	IVDMD	Tannins	Flavonoids	Glycosides
<i>A. officinalis</i>	Malvaceae	Root	900	591	198	180	29	57	800	–	+	–
<i>A. absinthium</i>	Asteraceae	Stem	890	659	497	100	16	121	300	+	+	+
<i>C. recutita</i>	Asteraceae	Flower	909	499	365	171	27	68	540	+	+	+
<i>F. vulgare</i>	Apiaceae	Seed	872	808	727	207	33	70	290	+	–	–
<i>F. officinalis</i>	Papaveraceae	Stem	880	420	342	156	25	114	610	+	+	–
<i>H. officinalis</i>	Lamiaceae	Stem	887	664	516	61	10	83	350	+	+	–
<i>I. helenium</i>	Asteraceae	Root	883	208	148	84	13	117	810	–	–	–
<i>M. sylvestris</i>	Malvaceae	Flower	896	739	227	225	36	79	600	+	+	–
<i>M. officinalis</i>	Lamiaceae	Stem	899	457	452	209	33	53	290	+	+	+
<i>P. hybridus</i>	Asteraceae	Root	892	407	283	112	18	126	690	–	–	+
<i>P. lanceolata</i>	Plantaginaceae	Leaf	893	446	308	116	19	56	470	+	–	+
<i>R. officinalis</i>	Lamiaceae	Leaf	887	537	461	75	12	108	650	+	+	+
<i>S. virgaurea</i>	Asteraceae	Stem	909	498	437	51	8	70	470	+	+	+
Herbmix			896	479	345	151	24	98	570			
Meadow hay			895	789	435	61	10	47	620			
Concentrate			880	152	56	215	34	68	805			

DM, dry matter; aNDF, neutral-detergent fiber; ADF, acidic-detergent fiber; CP, crude protein; N, nitrogen; IVDMD, in vitro dry-matter digestibility; +, the presence of phytochemicals; –, the absence of phytochemicals.

(aNDF) contents were analyzed as described by Van Soest et al. (1991) using an ANKOM 2000 (ANKOM Technology, Macedon, USA) with heat-stable α -amylase. The pH values of the batch cultures were measured using a pH meter (InoLab pH Level 1, Weilheim, Germany).

A liquid sample (4 ml) was collected for subsequent analysis of ammonia N. The concentration of ammonia N was determined in the fluid by the phenol-hypochlorite method (Broderick and Kang, 1980). The in vitro dry-matter digestibility was estimated from the difference in substrate weights before and after incubation. The contents of the batch cultures were transferred to centrifuge tubes and centrifuged at $3500 \times g$ for 10 min. The pellets were washed twice with 50 ml of distilled water, centrifuged and dried to a constant weight at 105°C for 3 d (Mellenberger et al., 1970). The volume of accumulated gas released from the recorded pressure or the volume of gas produced after 24 h was determined using a mechanical manometer fitted to a transducer (Premagas, Stará Turá, Slovak Republic), as described by Váradyová et al. (2005).

Qualitative phytochemical screening revealed the active compounds mainly tannins, flavonoids and glycosides (Table 1). For the qualitative tests of the presence of secondary plant metabolites, samples (20 g) of each of the ground medicinal plants were extracted in 200 ml of ethanol (ethanol/distilled water, 1:1) and stirred for 24 h at 20°C (Wencelová et al., 2014). This mixture was filtered and 100 ml of ethanol was added. Afterward, the mixture was filtered again and concentrated using a vacuum concentrator (Concentrator Plus 5305, Eppendorf). Chemical tests for the screening and identification of bioactive chemical constituents in the medicinal plants under study were carried out in extracts using standard procedures (Yadav and Agarwala, 2011; Sharma et al., 2012).

The SCFAs were analyzed by gas chromatography; 1 μl of sample was directly injected into a PerkinElmer Clarus 500 gas chromatograph (Perkin Elmer, Inc., Shelton, USA). For methane determination, 1 ml of gas was collected using an air-tight syringe (GASTIGHT Syringes, Hamilton Bonaduz AG, Switzerland) and injected into the gas chromatograph. The chromatograph was equipped with a flame ionization detection system for estimating methane and SCFA concentrations. The SCFAs were separated using a stainless steel packed column (2 m $\times$ 2 mm i.d.) with a phase composition of 10% Carbowax 20M-TPA + 1% H_3PO_4 on a 100/120 Supelcoport (Supelco, Bellefonte, USA), and peaks at retention times of 3.3, 4.7 and 6.8 min for acetic, propionic and *n*-butyric acid were quantified. The methane was analyzed using a 10% Squalane Chrom P mesh side 80/100, 2 m $\times$ 2 mm i.d. (Supelco, Bellefonte, USA) packed column, and a peak was observed at 0.33 min. The column oven temperature was programmed at 150°C for SCFA and 50°C for methane. Injector and

detector temperatures were programmed at 230°C for SCFA and 100°C for methane. The rates of gas flow were 40 ml/min for hydrogen and 400 ml/min for air in both analyses. The average flow of nitrogen carrier gas was set at 36 ψ for SCFA and 14 ψ for methane.

2.4. Parasitological techniques

Fecal samples were collected on D20, 27, 32, 50 and 60 post-inoculation and stored at 5°C in a cooling box until laboratory examination. A modified McMaster technique (Coles et al., 1992) with a sensitivity of 50 eggs per gram (EPG) of feces was used for the detection of strongylid eggs.

2.5. Hematological parameters

Samples of blood were collected from each animal at D0, D20, D32 and D50. Blood samples were collected from the jugular vein by using a 21-gauge needle and syringe placed into microtubes containing 1.6 mg/ml EDTA-K3 (Sarstedt AG & Co, Nümbrecht, Germany). Hematological parameters were determined immediately by an automated hematological analyzer (Abbott CELL-DYN 3700, Global Medical Instrumentation, Inc., Ramsey, USA).

Blood samples for sera were collected into 10 ml serum separator tubes (Sarstedt AG & Co, Nümbrecht, Germany) and centrifuged at 1200g for 10 min at room temperature. The sera were stored at -80°C until analysis.

2.6. Inflammatory response

The concentrations of serum amyloid A (SAA) and serum calprotectin were determined using commercial sheep ELISA kits (MyBioSource Ltd., San Diego, CA, USA). Undiluted serum samples were analyzed following the manufacturer's instructions. The SAA and calprotectin concentrations had coefficients of variation < 10 and $< 15\%$ (intra- and interassay), respectively. The sensitivities of the Elisa kits were 0.1 ng/ml for calprotectin and 0.1 $\mu\text{g/ml}$ for SAA. The detection range of the SAA kit was 3.12–100 $\mu\text{g/ml}$. All samples were analyzed in duplicate. The optical density of the samples was determined at 450 nm using an Apollo 11 LB913 Elisa absorbance reader (Berthold Technologies GmbH & Co. KG, Bad Wildbad, Germany). We analyzed both parameters at D32 and D50 and pooled the values for D32 and D50 with those for the control groups (C and C + Herb), because they did not differ significantly.

2.7. Antioxidant status

The total antioxidant capacity (TAC) of the serum was measured by an assay for ferric reducing antioxidant power described by Benzie and Strain (1996). A ferrous sulfate solution was used to create a standard curve, and the results were expressed in mmol Fe²⁺ formed per liter of sample. The activity of glutathione peroxidase (GPx) in the serum was assessed by the method described by Paglia and Valentine (1967), whereby one unit of GPx activity was defined as the amount of enzyme required to oxidize 1 μmol of NADPH to NADP⁺ per minute at 25 °C. The malondialdehyde (MDA) concentration in the serum was determined using the modified fluorometric method described by Jo and Ahn (1998).

2.8. Mineral status

Serum concentrations of Zn, Fe and Cu in the lambs were determined by flame atomic absorption spectrometry in an air-acetylene flame, with deuterium background correction, using an atomic absorption spectrophotometer AA-7000 (Shimadzu Co., Kyoto, Japan). Certified lyophilized human plasma, ClinCheck Control (Recipe, Munich, Germany), was used to determine the precision of the analytical procedure.

2.9. Statistical analysis

Statistical analysis used analysis of variance (GraphPad Prism, GraphPad Software, Inc., San Diego, CA, USA). Fermentation parameters and antioxidant status (i.e., Tables 2 and 7) were analyzed by one-way analysis of variance using Tukey's post hoc. Statistical analysis of body weight (Table 3), hematological parameters (Tables 4 and 5), inflammatory response (Table 6) and mineral status (Table 8) used analysis of variance as a repeated measures mixed model that represented four animal groups (Inf, Inf + Herb, C, C + Herb) and sampling days. Effects included in the model were treatment, time and interactions between treatment and time. Differences between infection groups (Inf) and treated group (Inf + Herb) and both control groups (C and C + Herb) were analyzed by two – way ANOVA with Bonferroni post-test. Student's *t*-tests were applied to assess the differences between the arithmetic means of egg output (EPG) on different sampling days for Inf and Inf + Herb (Fig. 1). The non-parametric Mann-Whitney *U* test was used if the data were not normally distributed. The level of statistical significance for all tests was set at

$P < 0.05$.

3. Results

3.1. In vitro fermentation parameters of the medicinal herbs and Herbmix

All fermentation parameters except pH were significantly influenced by the herbs ($P < 0.001$, Table 2). Ammonia N was highest for *A. absinthium*, *M. sylvestris* and *M. officinalis* and lowest for *P. hybridus*, *P. lanceolata*, *R. officinalis* and *S. virgaurea*. *A. officinalis*, *I. helenium* and *P. hybridus* produced the most gas. Methane production varied among the herb treatments. SCFA content and molar proportions of acetate, propionate, *n*-butyrate, isobutyrate, *n*-valerate and isovalerate also varied among the medicinal herbs. *A. officinalis*, *C. recutita*, *I. helenium* and *P. lanceolata* had the lowest Herbmix acetate:propionate ratio.

3.2. Parasitological status and mean live-weight gain of the lambs

EPG data from D32 were compared with data from subsequent sampling days, i.e. D32 with D50 and D32 with D60, for the two groups infected with *H. contortus* to measure the effect of Herbmix on egg output (Fig. 1). The arithmetic means of fecal egg counts for both groups increased until D32, but the egg counts were always lower in Inf + Herb than Inf after D32. The egg output in Inf + Herb was significantly lower on D50 and D60 than on D32 ($P < 0.05$ and < 0.01 , respectively). All lambs used in the experiment had similar initial body weights (BW_s; Table 3). The treatment and time affected BW_s ($P < 0.01$ and $P < 0.001$, respectively). BW_s differed by D32 ($P < 0.01$) in Inf vs. C and Inf vs. C + Herb and D50 ($P < 0.001$) in Inf vs. Herb, Inf vs. C and Inf vs. C + Herb. The treatment and treatment and time (Tr × T) interaction affected the mean cumulative live-weight gains (LWG_s; $P < 0.001$). LWG_s were higher in Inf + Herb, C and C + Herb than Inf by D32 ($P < 0.01$) and D50 ($P < 0.001$).

3.3. Hematological parameters of the lambs

The complete red blood cell (RBC) hemograms of each infected animal (Inf and Inf + Herb) identified clinical signs of haemonchosis such as anemia (Table 4). The RBC count, hemoglobin (HGB) level and hematocrit (HCT) were influenced by time, treatment and Tr × T interaction ($P < 0.05$, $P < 0.01$ and $P < 0.001$, respectively). RBC count, HGB level and HCT differ in Inf vs. C and Inf vs. C + Herb by

Table 2

pH ammonia N, gas, methane and short chain fatty acids of the selected herbs after 24 h of in vitro ruminal incubation (n = 9).

Item	pH	NH ₃ -N (mg/l)	Gas (ml/g DM)	CH ₄ (mmol)	SCFA (mmol/l)	Short chain fatty acids (molar proportion)						
						C ₂	C ₃	C ₄	isoC ₅	C ₆	isoC ₇	C ₂ /C ₃
<i>A. officinalis</i>	6.82	170e	180e	10.1e	54.6c	56.2a	19.8bc	14.4b	3.04bc	3.0c	2.73b	2.84a
<i>A. absinthium</i>	6.99	186f	150b	7.82c	43.2a	61.3b	16.6a	10.9a	3.73c	3.65c	2.75b	3.69b
<i>C. recutita</i>	6.91	176e	180e	8.54d	50.6b	59.7b	20.2bc	14.3b	2.10ab	2.48b	1.19a	2.96a
<i>F. vulgare</i>	6.85	132c	110a	5.90b	51.2b	59.5b	17.0a	12.3ab	3.82c	2.64b	4.03d	3.50b
<i>F. officinalis</i>	6.89	150d	160c	5.81ab	40.5a	62.6b	18.2b	15.1c	1.08a	1.45a	1.53a	3.44b
<i>H. officinalis</i>	7.02	144d	150b	4.61a	42.3a	64.5c	16.5a	14.6b	1.27a	1.40a	1.65a	3.91b
<i>I. helenium</i>	6.80	119b	170d	6.57bc	52.9b	55.9a	21.1c	17.2d	0.67a	2.14b	1.98ab	2.65a
<i>M. sylvestris</i>	6.83	184f	180e	9.13d	56.2c	61.7b	18.4b	11.0a	2.89b	2.49b	2.92b	3.35b
<i>M. officinalis</i>	6.95	193f	150b	7.61c	44.9ab	60.2b	17.8b	11.7a	3.30c	2.46b	3.59c	3.38b
<i>P. hybridus</i>	6.67	119a	180e	6.73c	51.2b	62.9b	17.9b	15.4c	0.99a	1.21a	1.38a	3.51b
<i>P. lanceolata</i>	7.01	124a	160c	7.59c	47.0b	56.9a	20.8c	13.5b	2.90b	2.01b	3.19c	2.74a
<i>R. officinalis</i>	6.98	111a	160c	5.21a	40.6a	65.4c	16.3a	14.2b	1.21a	1.19a	1.59a	4.01c
<i>S. virgaurea</i>	6.90	110a	150b	5.07a	44.5ab	66.9c	16.0a	12.6ab	0.80a	1.52a	1.35a	4.18c
Herbmix	6.95	130b	170d	6.17b	48.6b	63.4c	15.7a	13.9b	1.22a	1.33a	1.60a	2.97a
SEM	0.105	10.4	4.69	0.31	1.74	0.52	0.33	0.12	0.08	0.04	0.03	0.08
<i>P</i>	Ns	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001

SCFA, short chain fatty acid; C₂, acetate; C₃, propionate; C₄, *n*-butyrate; C₅, isobutyrate; C₆, *n*-valerate; C₇, isovalerate. Different letters within a column indicate significant differences at $P < 0.05$.

Table 3

Body weight (BW) and mean live-weight gain (LWG) of lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) (n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	SD	Significance of effects:			
							Treatment			Time
							Inf vs. Inf + Herb	Inf vs. C	Inf vs. C + Herb	
BW (kg)	0	12.2	12.5	12.7	12.5	1.638	Ns	Ns	Ns	***
	32	13.8	16.8	18.1	17.8	2.230	Ns	**	**	Ns
	50	14.4	17.9	19.3	19.6	2.548	***	***	***	
LWG (kg)	32	1.68	4.28	5.35	5.37	1.116	**	***	***	Ns
	50	2.22	5.42	6.58	6.73	1.273	***	***	***	**

Ns, not significant.

** $P < 0.01$.

*** $P < 0.001$.

D20 ($P < 0.001$), D32 ($P < 0.05$, $P < 0.01$ and $P < 0.001$, respectively) and D50 ($P < 0.001$). MCV differ in Inf vs. C by D32 ($P < 0.05$). The treatment and time affected white blood cell (WBC) levels (Table 5). WBC were significantly lower by D32 ($P < 0.01$) in Inf vs. C and Inf vs. C + Herb ($P < 0.05$) and in Inf vs. C by D50 ($P < 0.05$). Except of lymphocytes and eosinophils the time affected the differential counts of all WBC ($P < 0.05$).

3.4. Inflammatory response

Neither nematode infection nor Herbmix treatment affected the SAA concentrations, which ranged from 17.07 to 20.27 $\mu\text{g/ml}$ (Table 6). Mean serum calprotectin concentrations, however, were influenced by treatment, time and $\text{Tr} \times \text{T}$ interaction ($P < 0.05$ and $P < 0.001$, respectively) and differed significantly ($P < 0.001$) in Inf vs. Inf + Herb. Mean serum calprotectin concentration was 4.02 ng/ml in Inf + Herb by D32 but ranged from 2.58 to 2.88 ng/ml in the other groups. Calprotectin concentrations did not differ significantly among the other groups and sampling days.

Table 4

Red blood cell (RBC) parameters (HGB, hemoglobin; HCT, hematocrit; MCV, mean corpuscular volume) of lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) (n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	SD	Significance of effects:			
							Treatment			Time
							Inf vs. Inf + Herb	Inf vs. C	Inf vs. C + Herb	
RBC (n/l)	0	9.8	10.5	10.8	10.1	1.53	Ns	Ns	Ns	***
	20	6.0	6.8	10.0	10.4	1.07	Ns	***	***	**
	32	6.0	6.3	10.4	8.6	1.59	Ns	***	*	
	50	6.0	6.7	10.5	9.9	1.21	Ns	***	***	
HGB (g/l)	0	107	102.3	106.2	101.2	10.20	Ns	Ns	Ns	***
	20	59.9	64.5	92.1	92.8	9.35	Ns	***	***	***
	32	57.5	56.4	99.5	91.6	9.82	Ns	***	***	
	50	58.2	61.3	99.7	93.9	11.47	Ns	***	***	
HCT (l/l)	0	0.209	0.219	0.232	0.225	0.029	Ns	Ns	Ns	***
	20	0.144	0.158	0.220	0.232	0.025	Ns	***	***	*
	32	0.146	0.144	0.226	0.203	0.031	Ns	***	**	
	50	0.143	0.150	0.227	0.224	0.021	Ns	***	***	
MCV (fl)	0	21.6	21.1	21.5	22.2	1.59	Ns	Ns	Ns	*
	20	24.2	23.3	22.2	22.3	1.48	Ns	Ns	Ns	Ns
	32	24.7	23.1	21.8	23.0	2.07	Ns	*	Ns	
	50	24.2	22.8	21.7	22.9	2.13	Ns	Ns	Ns	

Ns, not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

3.5. Antioxidant status

TAC and serum GPx activity were significantly lower ($P < 0.008$ and $P < 0.005$, respectively) by D50 in both Inf and Inf + Herb than the control animals. Serum MDA concentration did not differ significantly between groups (Table 7).

3.6. Mineral status

The serum concentrations of Zn, Fe and Cu were influenced by treatment, time and $\text{Tr} \times \text{T}$ interaction ($P < 0.001$, Table 8). The Zn and Fe serum concentrations differed significantly in Inf vs. Inf + Herb, Inf vs. C and Inf vs. C + Herb by D32 ($P < 0.05$ and $P < 0.001$, respectively) and in Inf vs. Inf + Herb and Inf vs. C by D50 ($P < 0.001$). The serum concentrations of Zn and Fe were lower in both infected groups than in C and C + Herb. The serum Cu concentrations differ across treatment and time in Inf vs. Inf + Herb, Inf vs. C and Inf vs. C + Herb ($P < 0.05$, $P < 0.01$ and $P < 0.001$, respectively).

Table 5

Total leukocyte (WBC) and differential counts of lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) (n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	SD	Significance of effects:			
							Treatment			Time
							Inf vs. Inf + Herb	Inf vs. C	Inf vs. C + Herb	
WBC (g/l)	0	9.1	9.4	11.4	10.7	2.19	Ns	Ns	Ns	***
	20	7.7	8.3	9.3	9.1	1.44	Ns	Ns	Ns	Ns
	32	6.3	7.3	9.5	9.2	1.40	Ns	**	*	Ns
	50	5.5	5.6	8.5	7.4	1.37	Ns	*	Ns	Ns
Neutrophils (g/l)	0	3.1	4.1	4.1	4.5	1.60	Ns	Ns	Ns	***
	20	3.2	3.1	3.0	2.5	0.93	Ns	Ns	Ns	Ns
	32	2.3	3.0	3.1	3.6	0.99	Ns	Ns	Ns	Ns
	50	2.0	1.5	2.5	3.2	0.77	Ns	Ns	Ns	Ns
Lymphocytes (g/l)	0	1.6	1.8	2.5	2.7	1.19	Ns	Ns	Ns	Ns
	20	1.6	1.8	2.8	3.5	1.29	Ns	Ns	Ns	Ns
	32	1.3	0.9	2.9	2.7	1.01	Ns	Ns	Ns	Ns
	50	1.2	1.9	2.9	1.9	1.10	Ns	Ns	Ns	Ns
Monocytes (g/l)	0	3.5	2.9	3.8	2.4	1.16	Ns	Ns	Ns	***
	20	2.3	2.7	2.9	2.5	0.84	Ns	Ns	Ns	Ns
	32	2.1	2.3	2.9	2.3	0.72	Ns	Ns	Ns	Ns
	50	1.8	1.9	2.5	1.7	0.63	Ns	Ns	Ns	Ns
Eosinophils (g/l)	0	0.09	0.12	0.16	0.09	0.09	Ns	Ns	Ns	Ns
	20	0.09	0.07	0.32	0.15	0.22	Ns	Ns	Ns	Ns
	32	0.07	0.05	0.15	0.08	0.07	Ns	Ns	Ns	Ns
	50	0.04	0.07	0.13	0.26	0.13	Ns	Ns	Ns	Ns
Basophils (g/l)	0	0.92	0.61	0.89	1.0	0.49	Ns	Ns	Ns	***
	20	0.59	0.63	0.36	0.44	0.35	Ns	Ns	Ns	Ns
	32	0.50	0.55	0.44	0.46	0.38	Ns	Ns	Ns	Ns
	50	0.41	0.27	0.44	0.33	0.21	Ns	Ns	Ns	Ns

Ns, not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

4. Discussion

4.1. Effect of the medicinal herbs and Herbmix on the fermentation parameters in vitro

We tested the effects of 13 traditional medicinal herbs common in central Europe on fermentation parameters. Some herbs are often characterized as both poisonous and medicinal, having a beneficial or an adverse effect depending on the amount in the diet (Kabera et al., 2014). No adverse effects, however, were detected for the 13 herbs on the in vitro ruminal fermentations. The majority of phenolic compounds are released from forage fiber at ruminal pHs near 6.7 (Gaillard and Richards, 1975), but the pH of the ruminal fermentations in our experiment was not affected (i.e. 6.7–7.0). The concentration of ammonia N in vitro ranged from 110 to 186 mg/l and exceeded the maximal level for microbial growth, especially for *A. officinalis*, *A.*

absinthium, *C. recutita*, *M. sylvestris* and *M. officinalis*. Ammonia N concentrations for maximal microbial protein synthesis in vitro are normally 50–60 mg/l (Satter and Slyter, 1974; Mehrez et al., 1977). The in vivo studies of Hume et al. (1970) and Pisulewski et al. (1981) reported ammonia N concentrations of 88–130 and 27–100 mg/l, respectively. The ammonia N concentration in Herbmix was within the range of those previously reported (Pisulewski et al., 1981; ARC, 1984). The higher gas production for the roots of *A. officinalis*, *I. helenium* and *P. hybridus* were associated with higher digestibilities due to the rapidly digested portion of the substrate. These findings are likely associated with the lower content of non-digestible polysaccharides (Jung and Allen, 1995; Buxton and Redfearn, 1997) and with the higher contents of the more digestible root storage polysaccharides such as inulin (Schütz et al., 2006; Wencelová et al., 2015). The amount of fermentation gases decreased as the fiber content (aNDF and ADF) of the herbs increased (e.g. *A. absinthium*, *H. officinalis*, *F. vulgare*, *M.*

Table 6

Serum amyloid A (SAA) and calprotectin (Calpro) concentrations in lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) (n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	SD	Significance of effects:			
							Treatment			Time
							Inf vs. Inf + Herb	Inf vs. C	Inf vs. C + Herb	
SAA (µg/ml)	32	20.27	20.20	17.07	18.63	3.858	Ns	Ns	Ns	Ns
	50	17.68	20.54				Ns	Ns	Ns	Ns
Calpro (ng/ml)	32	2.58	4.02	2.69	2.62	0.361	Ns	Ns	Ns	*
	50	2.88	2.59				Ns	Ns	Ns	***

Ns, not significant.

* $P < 0.05$.

*** $P < 0.001$.

Table 7

Total antioxidant capacity (TAC) of serum, activity of glutathione peroxidase (GPx) and concentration of malondialdehyde (MDA) in the sera of lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) at D50 (mean $\pm$ SD, n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	P
TAS (mmol/l)	50	0.31 $\pm$ 0.04a	0.30 $\pm$ 0.03a	0.37 $\pm$ 0.03b	0.32 $\pm$ 0.01ab	0.008
GPx (U/ml)	50	0.07 $\pm$ 0.01a	0.08 $\pm$ 0.04a	0.16 $\pm$ 0.07b	0.14 $\pm$ 0.04ab	0.005
MDA (μ mol/l)	50	0.26 $\pm$ 0.04	0.25 $\pm$ 0.04	0.22 $\pm$ 0.05	0.23 $\pm$ 0.02	0.224

Different letters within a row indicate significant differences at $P < 0.05$.

officinalis and *R. officinalis*). Doane et al. (1997) reported a linear relationship between neutral detergent fiber disappearance and gas production in vitro. The acetate:propionate ratios produced in the herbal in vitro fermentations ranged from 2.65 (*I. helenium*) to 4.18 (*S. virgaurea*). The acetate:propionate ratio of Herbmix was within the ranges for meadow hay and wheat straw (Váradyová et al., 2005; Wencelová et al., 2015).

4.2. Effect of Herbmix on parasite egg counts and live-weight gains of the lambs

Many studies are exploring alternative options for managing parasitic infections in response to the rapid increase in anthelmintic resistance. Natural anthelmintics and dewormers have been studied for many decades. These dewormers have either the potential to decrease parasitic resistance because they likely use non-specific mechanisms to kill the worms or to increase the hosts' resistance to the worms, increasing their ability to expel adult nematodes (Mali and Mehta, 2008). The egg outputs of the two parasitized groups (Inf and Inf + Herb) in our study indicated that the lambs in the group fed Herbmix had a 45% lower mean EPG count on D50 than D32 and the EPG count decreased by 63% between D32 and D60, compared to 10 and 30% in infected group (Inf), respectively. Egg production by *H. contortus* females has been reported to remain at a high level up to D50 post-inoculation and then to decrease (Kerboeuf and Hubert, 1987; Borgsteede and Couwenberg, 1987). EPG counts in our study decreased rapidly in Inf + Herb after D32, which could be attributed to a possible deworming effect of the herbal treatment. Due to practical reasons there was not possible to prolong investigation up to D80 and evaluate also the effect of Herbmix on adult worms.

H. contortus has a strong negative impact on sheep production, causing low biological and economic efficiency of sheep flocks, indicated by retarded growth (Balic et al., 2000). Herbmix surprisingly had a significant effect on body-weight gain in Inf + Herb. Daily live-

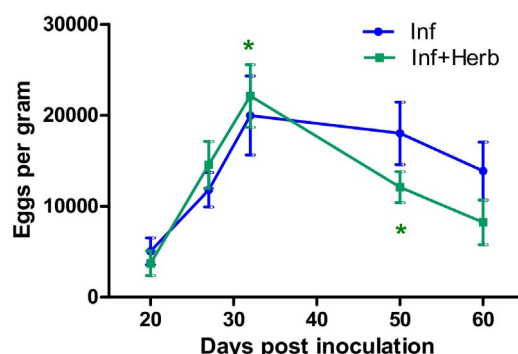


Fig. 1. Arithmetic mean fecal egg count for the groups of lambs infected with *H. contortus* (Inf and Inf + Herb). * D32 and D50 in the Inf + Herb group differed significantly ($P < 0.05$).

weight gains were significantly lower in Inf (44 g/d) than Inf + Herb (108.4 g/d) and were highest in C + Herb (134.6 g/d). The beneficial effects of plants against internal parasites may be due to one or more of the following factors. Tannins in the plants may increase the supply and absorption of digestible proteins by animals, which indirectly increases host resistance and resilience to infections by parasitic nematodes (Waller et al., 2001). Tannins may also have a direct anthelmintic effect, or tannins and/or metabolites in dung may have a direct effect on the viability of the free-living stages (Waller et al., 2001; Williams et al., 2014). Our data could be used to design an effective program for controlling parasitic nematodes of small ruminants. Additional study to confirm our findings, though, is necessary.

4.3. Effect of Herbmix on the hematological profile, inflammatory response, antioxidant status and mineral status of the lambs

The experimental infection with *H. contortus* caused clinical signs of haemonchosis such as anemia and hematological disturbances (Sharma

Table 8

Serum Zn, Fe and Cu concentrations in lambs infected with *H. contortus* (Inf), infected lambs supplemented with Herbmix (Inf + Herb), uninfected lambs (C) and uninfected lambs supplemented with Herbmix (C + Herb) (n = 6).

Item	Day	Inf	Inf + Herb	C	C + Herb	SD	Significance of effects:			
							Treatment			Time
							Inf vs. Inf + Herb	Inf vs. C	Inf vs. C + Herb	
Zn (mg/l)	0	0.509	0.479	0.460	0.419	0.02	Ns	*	***	***
	32	0.679	0.763	0.801	0.610	0.03	***	***	***	***
	50	0.703	0.841	0.794	0.718	0.03	***	***	Ns	***
Fe (mg/l)	0	1.945	1.651	1.733	1.884	0.22	Ns	Ns	Ns	***
	32	1.118	0.787	1.709	1.671	0.11	*	***	***	***
	50	1.360	2.466	2.413	2.583	0.26	***	***	***	***
Cu (mg/l)	0	0.810	0.740	0.789	0.699	0.04	**	Ns	***	***
	32	0.637	0.550	0.691	0.654	0.04	***	*	Ns	***
	50	0.682	0.680	0.772	0.680	0.03	Ns	***	Ns	***

Ns, not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

et al., 2000; Bordoloi et al., 2012). Both groups of lambs infected with *H. contortus* (Inf and Inf + Herb) had lower RBC, HGB and HCT values than the uninfected controls, as also reported for goat haemonchosis by Sharma et al. (2000) and Jan et al. (2015). Tannins from *Sericea Lespedeza* hay (Shaik et al., 2006), *Psidium guajava* and *Carissa spinarum* (Jan et al., 2015) positively influenced the RBC parameters of infected goats compared to infected goats without dietary tannins. Herbmix in our experiment, however, did not have a beneficial effect on the RBC parameters of the infected lambs, possibly due to either the composition of the lamb diet (Vargas-Magaña et al., 2013) or to the contents and properties of the plant secondary metabolites such as tannins (Min and Hart, 2003; Stürm et al., 2007), or to both. Total leukocyte levels were significantly lower by D32 and D50 in both Inf and Inf + Herb than C and C + Herb, consistent also with the lower neutrophil and lymphocyte levels. These lower levels could be ascribed to secondary bacterial infections in the infected lambs (Abramson and Mills, 1988) or to their different immune responses (Alba-Hurtado and Muñoz-Guzmán, 2013; Santos et al., 2014).

To the best of our knowledge, this study is the first in which the changes in the concentrations of acute-phase proteins (APPs), SAA and calprotectin have been associated with *H. contortus* infestation in sheep. The quantitative evaluation of serum SAA levels in lambs found no response from D32 to D50 either to the experimental infestation with *H. contortus* or to Herbmix treatment. The half-lives of most APPs are generally only 24–48 h (Cray et al., 2009; Piccione et al., 2012). Wells et al. (2013), however, demonstrated that APP (haptoglobin and SAA) concentrations were associated with sheep scab infestation for two months. SAA may thus not be a suitable biomarker of *H. contortus* infestation over long time periods. Examination of shorter time periods may be needed for using SAA as a biomarker of nematode infestation.

Calprotectin is an abundant protein in neutrophil granulocytes and monocytes. It can be released after leukocyte activation and/or endothelial adhesion (Stříž and Trebichavský, 2004). Calprotectin has antimicrobial properties and has induced apoptosis in all cell types tested (human, animal, normal and malignant) (Vaos et al., 2013). The calprotectin concentration in our experiment was higher only in Inf + Herb by D32. No differences were detected between groups by D50. Treatment of infected lambs with Herbmix may have strongly stimulated the immune system, which helped to manage the nematode infestation in this group of animals. Chemical components of Herbmix, however, may have directly influenced the nematodes, suggested by the significantly lower EPG counts at D50 and D60 in Inf + Herb (Fig. 1).

The analysis of antioxidant parameters identified significantly lower serum TAC and GPx activity in both infected groups. The serum MDA level, as a marker of lipid peroxidation, tended to be higher in the infected than the uninfected groups. Nanev et al. (2012) reported similar findings; MDA production in plasma was higher and the activity of blood GPx was lower in lambs infected with *H. contortus*. Our results indicated that nematode infection in lambs is associated with oxidative stress and affects the antioxidant status of animals. Medicinal plants containing phenolic compounds have been reported to possess strong antioxidant properties, and the compounds may help to protect cells against oxidative damage (Kasote et al., 2015). Herbmix, however, did not have a beneficial effect on the serum antioxidant parameters in the lambs in our experiment.

The lower serum Zn, Fe and Cu levels of the infected lambs (Inf) suggest a disturbance in mineral metabolism caused by *H. contortus*. We should stress that the serum mineral status of the C + Herb lambs was determined at the beginning the experiment, and this trend was followed throughout the experimental period. Blood or plasma Cu levels and hepatic Cu and Zn concentrations are lower in ruminants infected with gastrointestinal nematodes, including *H. contortus* (Schafer et al., 2015). Increases in pH in animal gastrointestinal tracts caused by nematode infections decreases the availability of trace elements due to their reduced absorption from the intestine (Bank et al., 1990). Poor mineral status affects the immune system of the gut

mucosa and can change systematic immune responses in prolonged parasitic infections. Herbal treatment may improve the Zn and Fe status of infected lambs by a beneficial effect of Herbmix on the serum mineral levels of infected lambs, regardless of plasma Cu level.

5. Conclusion

Our study of lambs infected with the gastrointestinal nematode *H. contortus* underlined the potential value of the medicinal herbs to decrease egg output of parasitic infection of the digestive tract. The combination of these medicinal herbs belonging to different botanical families contributed to slowing the dynamics of infection and improved the production indicator of the lambs. Herbmix represents a promising herbal mixture of medicinal plants for controlling nematode infections, and its use is compatible with the principles of organic farming.

Conflict of interest

The authors declare that there is no conflict of interest.

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References

- Association of Official Analytical Chemists, 1990. Official Methods of Analysis, 15th ed. AOAC, Arlington.
- ARC, 1984. The Nutrient Requirements of Ruminant Livestock. Supplement No.1. CAB International, Wallingford, UK.
- Abramson, J.S., Mills, E.L., 1988. Depression of neutrophil function induced by viruses and its role in secondary microbial infections. Rev. Infect. Dis. 10, 326–341.
- Alba-Hurtado, F., Muñoz-Guzmán, M.A., 2013. Immune responses associated with resistance to haemonchosis in sheep. Biomed. Res. Int. 2013, 162158.
- Balic, A., Bowles, V.M., Meeusen, E.N.T., 2000. The immunobiology of gastrointestinal nematode infection in ruminants. Adv. Parasitol. 45, 181–241.
- Bank, K.S., Familton, A.S., Sykes, A.R., 1990. Effect of ostertagiasis on copper status in sheep: a study involving use of copper oxide wire particles. Res. Vet. Sci. 49, 306–314.
- Benzie, I.F.F., Strain, J.J., 1996. The ferric reducing ability of plasma (FRAP) as a measure of "Antioxidant Power": the FRAP assay. Anal. Biochem. 239, 70–76.
- Bhatta, R., Baruah, L., Saravanan, M., Suresh, K.P., Sampath, K.T., 2013. Effect of medicinal and aromatic plants on rumen fermentation, protozoa population and methanogenesis in vitro. J. Anim. Physiol. Anim. Nutr. 97, 446–456.
- Bordoloi, G., Jas, R., Ghosh, J.D., 2012. Changes in the haemato-biochemical pattern due to experimentally induced haemonchosis in Sahabadi sheep. J. Parasite Dis. 36, 101–105.
- Borgsteede, H.M., Couwenberg, T., 1987. Changes in LC50 in an in vitro egg development assay during the patent period of *Haemonchus contortus* in sheep. Res. Vet. Sci. 42, 413–414.
- Broderick, G.A., Kang, J.H., 1980. Automated simultaneous determination of ammonia and total amino acids in ruminal fluid and in vitro media. J. Dairy Sci. 63, 64–75.
- Buxton, D.R., Redfearn, D.D., 1997. Plant limitations to fiber digestion and utilization. J. Nutr. 127, 814S–818S.
- Coles, G.C., Bauer, C., Borgsteede, F.H.M., Geerts, S., Klei, T.R., Taylor, M.A., Waller, P.J., 1992. World Association for the Advancement of Veterinary Parasitology (W.A.A.V.P) methods for the detection of anthelmintic resistance in nematodes of veterinary importance. Vet. Parasitol. 44, 35–44.
- Coop, R.L., Kyriazakis, I., 2001. Influence of host nutrition on the development and consequences of nematode parasitism in ruminants. Trends Parasitol. 17, 325–330.
- Cray, C., Zaias, J., Altman, N.H., 2009. Acute phase response in animals: a review. Comp. Med. 59, 517–526.
- Doane, P.H., Schofield, P., Pell, A.N., 1997. Neutral detergent fibre disappearance and gas and volatile fatty acids production during the in vitro fermentation of six forages. J. Anim. Sci. 75, 3342–3352.
- Fraisse, D., Carnat, A., Viala, D., Pradel, P., Besle, J.M., Coulon, J.B., Felgines, C., Lamaison, J.L., 2007. Polyphenolic composition of a permanent pasture: variations related to the period of harvesting. J. Sci. Food Agric. 87, 2427–2435.
- Gaillard, B.D., Richards, G.N., 1975. Presence of soluble lignin-carbohydrate complexes in the bovine rumen. Carbohydr. Res. 42, 135–145.
- Githiori, J.B., Athanasiadou, S., Thamsborg, S.M., 2006. Use of plants in novel approaches for control of gastrointestinal helminths in livestock with emphasis on small ruminants. Vet. Parasitol. 139, 308–320.
- Hammond, J.A., Fielding, D., Bishop, S.C., 1997. Prospects for plant anthelmintics in tropical veterinary medicine. Vet. Res. Commun. 21, 213–228.

- Hosoda, K., Kuramoto, K., Eruden, B., Nishida, T., Shioya, S., 2006. The effect of three herbs as feed supplements on blood metabolites, hormones, antioxidant activity, IgG concentration, and ruminal fermentation in Holstein steers. *Asian-Aust. J. Anim. Sci.* 19, 35–41.
- Hoste, H., Jackson, F., Athanasiadou, S., Thamsborg, S.M., Hoskin, S.O., 2006. The effects of tannin-rich plants on parasitic nematodes in ruminants. *Trends Parasitol.* 22, 253–261.
- Hoste, H., Manolaraki, F., Brunet, S., Arroyo López, C., Martínez-Ortiz de Montellano, C., Sotiraki, S., Torres Acosta, F., 2011. The anthelmintic properties of tannin-rich legume forages: from knowledge to exploitation in farm conditions. In: Ranilla, M.J., Carro, M.D., Ben Salem, H., Morand-Fehr, P. (Eds.), *Challenging Strategies to Promote the Sheep and Goat Sector in the Current Global Context*. CIHEAM/CSIC/Universidad de León/FAO, Zaragoza, pp. 295–304.
- Hughes, S., Kelly, P., 2006. Interaction of malnutrition and immune impairment: with specific reference to immunity against parasites. *Parasite Immunol.* 28, 577–588.
- Hume, I.D., Moir, R.J., Somers, M., 1970. Synthesis of microbial protein in the rumen. I. Influence of the level of nitrogen intake. *Aust. J. Agric. Res.* 21, 283–296.
- Jan, O.Q., Kamili, N., Ashraf, A., Iqbal, A., Sharma, R.K., Rastogi, A., 2015. Haematobiochemical parameters of goats fed tannin rich Psidium guajava and Carissa spinarum against *Haemonchus contortus* infection in India. *J. Parasite Dis.* 39, 41–48.
- Jo, C., Ahn, D.U., 1998. Fluorometric analysis of 2-thiobarbituric acid reactive substances in turkey. *Poult. Sci.* 77, 475–480.
- Jung, H.G., Allen, M.S., 1995. Characteristics of plant cell walls affecting intake and digestibility of forages by ruminants. *J. Anim. Sci.* 73, 2774–2790.
- Kabera, J.N., Semana, E., Mussa, A.R., He, X., 2014. Plant secondary metabolites: biosynthesis, classification, function and pharmacological properties. *J. Pharm. Pharmacol.* 2, 377–392.
- Karaskova, K., Suchy, P., Strakova, E., 2015. Current use of phytogetic feed additives in animal nutrition: a review. *Czech J. Anim. Sci.* 60, 521–530.
- Kasote, D.M., Katyare, S.S., Hegde, M.V., Bae, H., 2015. Significance of antioxidant potential of plants and its relevance to therapeutic application. *Int. J. Biol. Sci.* 11, 982–991.
- Kerboeuf, D., Hubert, J., 1987. Changes in the response of *Haemonchus contortus* eggs to the ovidal activity of thiabendazole during the course of infection. *Ann. Rech. Vet.* 18, 365–370.
- Kresánek Jr., J., Kresánek, J., 2008. *Atlas of Medicinal Plants and Berries*, fourth ed. Osveta, Slovakia.
- Lans, C., Turner, N., Khan, T., Brauer, G., Boepple, W., 2007. Ethnoveterinary medicines used for ruminants in British Columbia, Canada. *J. Ethnobiol. Ethnomed.* 3, 11.
- Machado, V., Da Silva, A.S., Schafer, A.S., Aires, A.R., Tonin, A.A., Oliveira, C.B., Hermes, C.L., Almeida, T.C., Moresco, R.N., Stefani, L.M., Lopes, S.T., Mendes, R.E., Schwert, C.I., Leal, M.L., 2014. Relationship between oxidative stress and pathological findings in abomasum of infected lambs by *Haemonchus contortus*. *Pathol. Res. Pract.* 210, 812–817.
- Mali, R.G., Mehta, A.A., 2008. A review on anthelmintic plants. *Nat. Prod. Rad.* 7, 466–475.
- McDougall, E.I., 1948. Studies on ruminant saliva: I. The composition and output of sheep's saliva. *Biochem. J.* 43, 99–109.
- Mehrez, A.Z., Orskov, E.R., McDonald, I., 1977. Rates of rumen fermentation in relation to ammonia concentration. *Br. J. Nutr.* 38, 437–443.
- Mellenberger, R.W., Satter, L.D., Millett, M.A., Baker, A.J., 1970. An in vitro technique for estimating digestibility of treated and untreated wood. *J. Anim. Sci.* 30, 1005–1011.
- Min, B.R., Hart, S.P., 2003. Tannins for suppression of internal parasites. *J. Anim. Sci.* 81, E102–E109.
- Nanev, V., Gabrashanska, M., Vladov, I., 2012. Impact of *Haemonchus contortus* (Nematoda) infection on the oxidant/antioxidant status in lambs. *Ecologica* 19, 426–429.
- Niezen, J.H., Waghorn, T.S., Charleston, W.A.G., Waghorn, G.C., 1995. Growth and gastrointestinal nematode parasitism in lambs grazing either lucerne (*Medicago sativa*) or sulla (*Hedysarum coronarium*) which contains condensed tannins. *J. Agric. Sci.* 125, 281–289.
- Paglia, D.E., Valentine, W.N., 1967. Studies on the quantitative and qualitative characterization of erythrocyte glutathione peroxidase. *J. Lab. Clin. Med.* 70, 158–169.
- Patra, A.K., Saxena, J., 2009. Dietary phytochemicals as rumen modifiers: a review of the effects on microbial populations. *Antonie van Leeuwenhoek* 96, 363–375.
- Patra, A.K., 2007. Nutritional management in organic livestock farming for improved ruminant health and production – an overview. *Livest. Res. Rural Dev.* 19 (3).
- Piccione, G., Casella, S., Giannetto, C., Giudice, E., Fazio, F., 2012. Utility of acute phase proteins as biomarkers of transport stress in ewes. *Small Rumin. Res.* 107, 167–171.
- Pisulewski, P.M., Okorie, A.U., Buttery, P.J., Haresign, W., Lewis, D., 1981. Ammonia concentration and protein synthesis in the rumen. *J. Sci. Food Agric.* 32, 759–766.
- Santos, M.C., Xavier, J.K., Amarante, M.R., Bassetto, C.C., Amarante, A.F., 2014. Immune response to *Haemonchus contortus* and *Haemonchus placei* in sheep and its role on parasite specificity. *Vet. Parasitol.* 16 (203), 127–138.
- Satter, L.D., Slyter, L.L., 1974. Effect of ammonia concentration on rumen microbial protein production in vitro. *Br. J. Nutr.* 32, 199–208.
- Schütz, K., Muks, E., Carle, R., Schieber, A., 2006. Separation and quantification of inulin in selected artichoke (*Cynara scolymus* L.) cultivars and dandelion (*Taraxacum officinale* WEB. ex WIGG.) roots by high-performance anion exchange chromatography with pulsed amperometric detection. *Biomed. Chromatogr.* 20, 1295–1303.
- Schafer, A.S., Leal, M.L.R., Molento, M.B., Aires, A.R., Duarte, M.M.M.F., Carvalho, F.B., Tonin, A.A., Schmidt, L., Flores, E.M.M., Franca, R.T., Grando, T.H., Minho, A.P., Krause, A., Antoniazzi, A.Q., Lopes, S.T.A., 2015. Immune response of lambs experimentally infected with *Haemonchus contortus* and parenterally treated with a combination of zinc and copper. *Small Rumin. Res.* 123, 183–188.
- Shaik, S.A., Terrill, T.H., Miller, J.E., Kouakou, B., Kannan, G., Kaplan, R.M., Burke, J.M., Mosjidis, J.A., 2006. *Sericea Lespedeza* hay as a natural deworming agent against gastrointestinal nematode infection in goats. *Vet. Parasitol.* 139, 150–157.
- Sharma, D.K., Chauhan, P.P.S., Agrawal, R.D., 2000. Haematological changes in experimental haemonchosis in Barbari goats. *J. Anim. Sci.* 70, 353–355.
- Sharma, A.K., Gangwar, M., Tilak, R., Nath, G., Sinha, A.S.K., Tripathi, Y.B., Kumar, D., 2012. Comparative in vitro antimicrobial and phytochemical evaluation of methanolic extract of root: stem and leaf of *Jatropha curcas* Linn. *Pharmacogn. J.* 4, 34–40.
- Stürm, C.D., Tiemann, T.T., Lascano, C.E., Kreuzer, M., Hess, H.D., 2007. Nutrient composition and in vitro ruminal fermentation of tropical legume mixtures with contrasting tannin contents. *Anim. Feed Sci. Technol.* 138, 29–46.
- Stříž, I., Trebichavský, I., 2004. Calprotectin – a pleiotropic molecule in acute and chronic inflammation. *Physiol. Res.* 53, 245–253.
- Váradyová, Z., Baran, M., Zelenák, I., 2005. Comparison of two in vitro fermentation gas production methods using both rumen fluid and faecal inoculum from sheep. *Anim. Feed Sci. Technol.* 123–124, 81–94.
- Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fiber neutral detergent fiber, and non-starch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74, 3583–3597.
- Vaas, G., Kostakis, I.D., Zavras, N., Chatzemichael, A., 2013. The Role of Calprotectin in pediatric disease. *Biomed. Res. Int.* 2013, 542363.
- Vargas-Magaña, J.J., Aguilar-Caballero, A.J., Torres-Acosta, J.F., Sandoval-Castro, C.A., Hoste, H., Capetillo-Leal, C.M., 2013. Tropical tannin-rich fodder intake modifies saliva-binding capacity in growing sheep. *Animal* 7, 1921–1924.
- Viegi, L., Pieroni, A., Guarrera, P.M., Vangelisti, R., 2003. A review of plants used in folk veterinary medicine in Italy as basis for a databank. *J. Ethnopharmacol.* 89, 221–244.
- Villalba, J.J., Miller, J., Ungar, E.D., Landau, S.Y., Glendinning, J., 2014. Ruminant self-medication against gastrointestinal nematodes: evidence, mechanism, and origins. *Parasite* 21, 31.
- Waller, P.J., Thamsborg, S.M., 2004. Nematode control in 'green' ruminant production systems. *Trends Parasitol.* 20, 493–497.
- Waller, P.J., Bernes, G., Thamsborg, S.M., Sukura, A., Richter, S.H., Ingebrigtsen, K., Höglund, J., 2001. Plants as de-worming agents of livestock in the nordic countries: historical perspective, popular beliefs and prospects for the future. *Acta Vet. Scand.* 42, 31–44.
- Wells, B., Innocent, G.T., Eckersall, P.D., McCulloch, E., Nisbet, A.J., Burgess, S.T., 2013. Two major ruminant acute phase proteins, haptoglobin and serum amyloid A: as serum biomarkers during active sheep scab infestation. *Vet. Res.* 44, 103–114.
- Wencelová, M., Váradyová, Z., Mihalíková, K., Jalč, D., Kišidayová, S., 2014. Effects of selected medicinal plants on rumen fermentation in a high-concentrate diet in vitro. *J. Anim. Plant Sci.* 24, 1388–1395.
- Wencelová, M., Váradyová, Z., Mihalíková, K., Čobanová, K., Plachá, I., Pristaš, P., Jalč, D., Kišidayová, S., 2015. Rumen fermentation pattern, lipid metabolism and the microbial community of sheep fed a high-concentrate diet supplemented with a mix of medicinal plants. *Small Rumin. Res.* 125, 64–72.
- Williams, A.R., Ropiak, H.M., Frygas, C., Desrués, O., Mueller-Harvey, I., Thamsborg, S.M., 2014. Assessment of the anthelmintic activity of medicinal plant extracts and purified condensed tannins against free-living and parasitic stages of *Oesophagostomum dentatum*. *Parasite Vectors* 7, 518.
- Wink, M., 2012. Medicinal plants: a source of anti-parasitic secondary metabolites. *Molecules* 17, 12771–12791.
- Yadav, R.N.S., Agarwala, M., 2011. Phytochemical analysis of some medicinal plants. *J. Phytol.* 3, 10–14.
- Yadav, P., Kumar, A., Vihan, V.S., Mahour, K., 2010. In-vitro adulticidal activity of various plant extracts against *H. contortus*. *Asian J. Exp. Biol. Sci.* 1, 975–978.